

Quiz-1 – EE101T (S1): Introduction to Electrical Systems and Electronics

Q1: Fill the third column of the following table selecting the correct option from fourth column. **(2-marks)**

Sr. No.	Electrical Quantity	Unit	Options
1	Current	ampere	watt-hour, ampere-hour, joule/second, ohm-m, ampere, volt, siemens/m, ampere/m ² , ampere/m, ampere/m ³ ,
2	Energy	watt-hour	
3	Power	joule/second	
4	Node-voltage	volt	
5	Conductivity	siemens/m	
6	Charge stored in battery	ampere-hour	
7	Resistivity	ohm-m	
8	Current-density	ampere/m ²	

0.25 marks for each correct response. No partial marking

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Q2: In the following table, some electrical items along with their rated specifications are mentioned. For each of these items, determine the quantity mentioned in fourth column. **(2-marks)**

Sr. No.	Electrical Item	Specifications	Determine
1	Electric-Iron	1000 W, 220 V, 50 Hz	Current drawn from supply
2	Phone-charger	20 W, 9 V output	Current delivered to the phone
3	Ola EV battery	2.4 kWh, 72 V, rated-charging current: 7 A	Charging-duration for 0% to 100%
4	Smart-watch cell	250 mAh, 3.7 V, average discharge current: 5 mA	Usage-time (hour) for 100% charge

0.25 marks for each correct procedure and 0.25 marks for value. Allow variation of -5% to +5% of the correct answer.

No partial marking

$$1. \quad P = 1000 \text{ W}, \quad P = V \cdot I \rightarrow I = \frac{P}{V} = \frac{1000}{220} = 4.54 \text{ A}$$

$$2. \quad P = 20 \text{ W}, \quad I = \frac{P}{V} = \frac{20}{9} = 2.22 \text{ A}$$

$$3. \quad \text{Energy-storage Capacity} = 2.4 \text{ kWh} = 2400 \text{ Wh} = 2400 \text{ VAh}$$

Nominal voltage = 72 V

$$\therefore \text{Charge-storage Capacity} = \frac{2400 \text{ VAh}}{72} = 33.3 \text{ Ah}$$

$$\text{Charge-duration} = \frac{33.33 \text{ Ah}}{7 \text{ A (rated-current)}} = 4.76 \text{ hour}$$

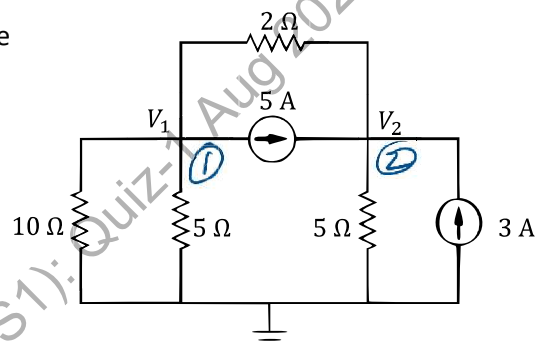
(0% to 100%)
= 4 hour 46 minute

$$4. \quad \text{Charge storage capacity} = 250 \text{ mAh}$$

$$\text{discharge-duration} = \frac{250 \text{ mAh}}{5 \text{ mA}} = 50 \text{ hour}$$

= 2 day 2 hour

Q3: Obtain the node-voltages V_1 and V_2 using node-method. Determine the value of power-absorbed by all the elements. [Note: You are not allowed to use the formula " $i^2 R$ " for the resistors.] (3-marks)



$$\begin{aligned} \textcircled{1} & \begin{bmatrix} \frac{1}{10} + \frac{1}{5} + \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} + \frac{1}{5} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -5 \\ 5+3 \end{bmatrix} \\ \textcircled{2} & \begin{bmatrix} 0.8 & -0.5 \\ -0.5 & 0.7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -5 \\ 8 \end{bmatrix} \end{aligned}$$

1-mark for procedure. Matrix-based approach is not mandatory.

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0.8 & -0.5 \\ -0.5 & 0.7 \end{bmatrix}^{-1} \begin{bmatrix} -5 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1.613 \\ 12.58 \end{bmatrix}$$

0.5 marks for correct value of V_1
0.5 marks for correct value of V_2

power-absorbed calculations

element

$$10\Omega: P_{10\Omega} = V_{10\Omega} \cdot I_{10\Omega} = (1.61) \left(\frac{1.61}{10} \right) = 0.26 \text{ W}$$

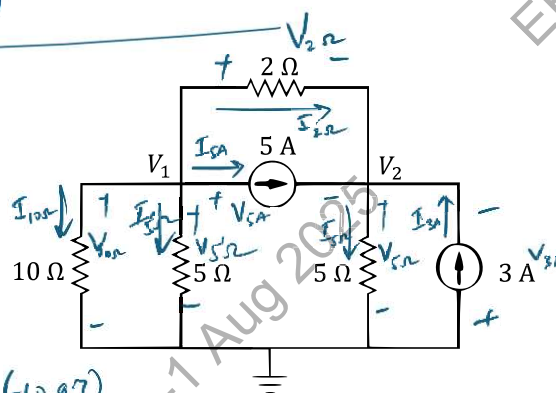
$$5\Omega: P_{5\Omega} = V_{5\Omega} \cdot I_{5\Omega} = (1.61) \left(\frac{1.61}{5} \right) = 0.52 \text{ W}$$

$$2\Omega: P_{2\Omega} = V_{2\Omega} \cdot I_{2\Omega} = \frac{(V_1 - V_2)(V_1 - V_2)}{2} = \frac{(-10.97)(-10.97)}{2} = +60.17 \text{ W}$$

$$5\Omega: P_{5\Omega} = V_{5\Omega} \cdot I_{5\Omega} = (12.58) \left(\frac{12.58}{5} \right) = 31.65 \text{ W}$$

$$5A: P_{5A} = V_{5A} \cdot I_{5A} = (V_1 - V_2)(5) = (-10.97)(5) = -54.85 \text{ W}$$

$$3A: P_{3A} = V_{3A} \cdot I_{3A} = (-V_2)(3) = (-12.58)(3) = -37.74 \text{ W}$$



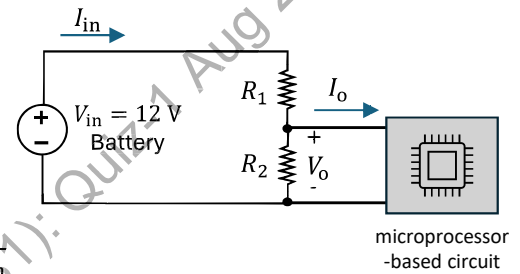
Flat deduct 0.5 marks if students have used formula other than $V \cdot I$ for any of the resistors

Deduct 0.25 marks for each incorrect value of power-absorbed

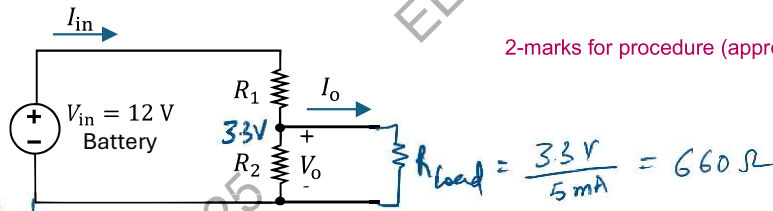
Q4: A certain microprocessor-based circuit requires 3.3 V supply. At full-load condition, this circuit may draw up to 5 mA current. You only have 12 V battery available with you, and you are required to use a resistor voltage-divider to obtain 3.3 V using 12 V battery. The schematic representation is given in the figure. (4-marks)

(a) Determine the values of R_1 and R_2 that give $V_0 = 3.3$ V at full-load.

(b) Determine the efficiency of this voltage-conversion system, i.e. $\frac{V_0 I_0}{V_{in} I_{in}}$

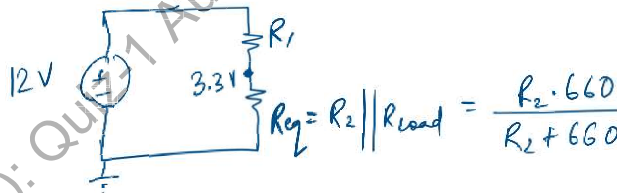


Model at full-load



2-marks for procedure (approach)

This is equivalent to the following:



Now, we know that in the above voltage divider circuit:

$$3.3 = \frac{R_{eq}}{R_{eq} + R_1} \cdot 12 = \frac{\left(\frac{660 \cdot R_2}{R_2 + 660}\right)}{\left(\frac{660 \cdot R_2}{R_2 + 660}\right) + R_1} \cdot 12$$

$$\therefore \frac{660R_2 + R_1R_2 + 660R_1}{660R_2} = \frac{12}{3.3} = 3.636$$

$$\therefore 660R_2 + R_1R_2 + 660R_1 = 2400R_2$$

$$\therefore R_1R_2 + 660R_1 = 1740R_2$$

$$\therefore R_2 = \frac{660R_1}{(1740 - R_1)}$$

1-mark for correct values of R_1 and R_2 .

Any pair of values of R_1 and R_2 that satisfy this equation are correct.

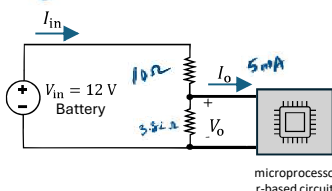
Assume $R_1 = 10 \Omega \Rightarrow 10 \cdot R_2 + 6600 = 1740R_2 \Rightarrow R_2 = 3.82 \Omega$

OR Assume $R_1 = 100 \Omega \Rightarrow 100 \cdot R_2 + 66000 = 1740R_2 \Rightarrow R_2 = 40.2 \Omega$

OR Assume $R_1 = 1000 \Omega \Rightarrow 1000R_2 + 660000 = 1740R_2 \Rightarrow R_2 = 892 \Omega$

OR Assume $R_1 = 2000 \Omega \Rightarrow 2000R_2 + 1320000 = 1740R_2 \Rightarrow R_2 = -5047 \Omega$ (? Whatt!!)

Choosing $R_1 = 10 \Omega$ & $R_2 = 3.82 \Omega$



$$S_{in} = \frac{12 - 3.3}{10} = 0.87 \text{ A}$$

$$I_0 = 5 \text{ mA}$$

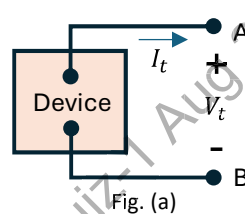
$$\eta = \frac{3.3 \times 0.005}{12 \times 0.87} = 0.00158 = 0.158 \%$$

1-mark for correct value of efficiency.

Q5: The voltage and current were measured at the terminals of the device shown in Fig.(a). The results are tabulated in Fig.(b). **(4-marks)**

(a) Determine if this device is linear or not.

(b) Obtain the Thevenin's equivalent network of this device.



V_t (V)	I_t (A)
90	1
80	2
70	3

Fig. (b)

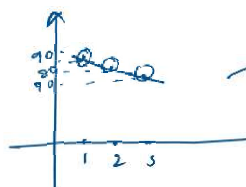
2-mark for approach

1-marks for correct response

(a) $I_{t1} = 1A \rightarrow V_{t1} = 90$

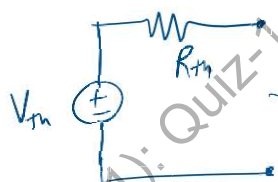
$I_{t2} = 2A \rightarrow V_{t2} = 80$

$I_{t1} + I_{t2} = 3A \rightarrow 70V \neq (90 + 80) \rightarrow \underline{\underline{\text{Not-linear}}}$



from plot (V-I-relationship) also, we can see that this device doesn't follow homogeneity.
 $\rightarrow$ Not-linear.

(b)



$V_{th} = V_{AB} \Big|_{I_t=0} = 100V$ (extrapolated from the values)

0.5-marks for correct value of V_{th}

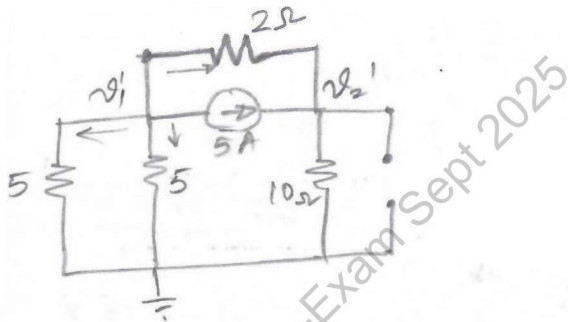
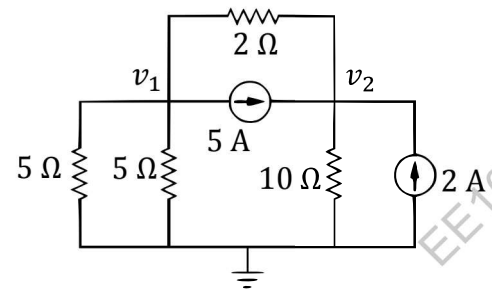
$V_t = V_{th} - I_t \cdot R_{th}$

put any of the values of V_t & I_t in the above equation to get R_{th} :

$70 = 100 - 3 \cdot R_{th} \Rightarrow R_{th} = 10\Omega$

0.5-marks for correct value of R_{th}

Q1: Obtain the values of node-voltages v_1 and v_2 using superposition approach. Strictly follow the passive sign-convention. **(5-marks)**



$$\frac{v_1'}{5} + \frac{v_1'}{5} + \frac{v_1' - v_2'}{2} = 0$$

$$5 + \frac{v_1' - v_2'}{2} = \frac{v_2'}{10}$$

1-mark for procedure and only one source active - Means superposition approach

$$\therefore 0.9v_1' - 0.5v_2' = -5$$

$$-0.5v_1' + 0.6v_2' = 5$$

$$\begin{bmatrix} 0.9 & -0.5 \\ -0.5 & 0.6 \end{bmatrix} \begin{bmatrix} v_1' \\ v_2' \end{bmatrix} = \begin{bmatrix} -5 \\ 5 \end{bmatrix}$$

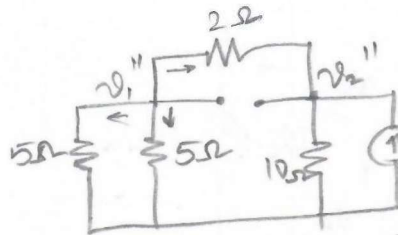
$$\begin{bmatrix} v_1' \\ v_2' \end{bmatrix} = \begin{bmatrix} -1.724 \\ 6.897 \end{bmatrix}$$

0.5 + 0.5 for each correct value of v_1' and v_2'

According to superposition,

$$v_1 = v_1' + v_1'' = 1.724 \text{ V}$$

$$v_2 = v_2' + v_2'' = 13.104 \text{ V}$$



$$\frac{v_1''}{5} + \frac{v_1''}{5} + \frac{v_1'' - v_2''}{2} = 0$$

$$-0.5v_1'' + 0.6v_2'' = 2$$

1-mark for procedure and only one source active - Means superposition approach

$$\therefore 0.9v_1'' - 0.5v_2'' = 0$$

$$-0.5v_1'' + 0.6v_2'' = 2$$

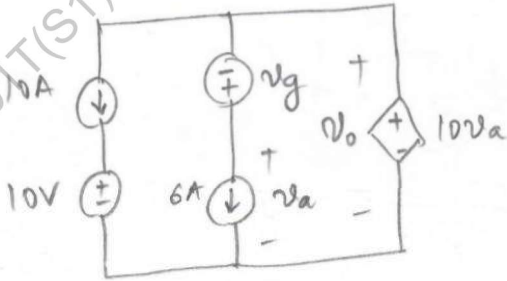
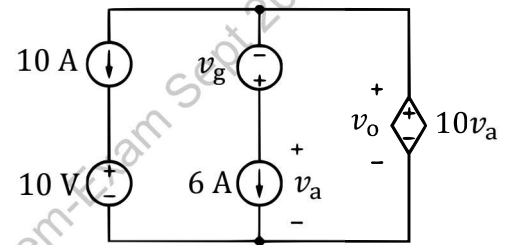
$$\begin{bmatrix} 0.9 & -0.5 \\ -0.5 & 0.6 \end{bmatrix} \begin{bmatrix} v_1'' \\ v_2'' \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} v_1'' \\ v_2'' \end{bmatrix} = \begin{bmatrix} 3.448 \\ 6.207 \end{bmatrix}$$

0.5 + 0.5 for each correct value of v_1'' and v_2''

0.5 + 0.5 for each correct value of v_1 and v_2

Q2: Obtain the value of 'power-absorbed' by each of the elements in the given circuit. The value of v_o is 20 V. Strictly follow the passive sign-convention. **(5-marks)**

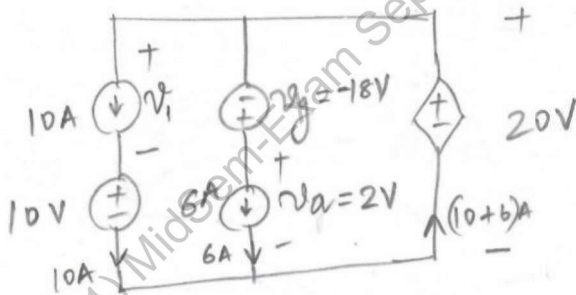


according to KVL,

$$v_o = 10v_a = v_a - v_g$$

Since $v_o = 20 \text{ V} \Rightarrow \underline{v_a = 2 \text{ V}}$

$$\therefore 20 = 2 - v_g \Rightarrow v_g = -18 \text{ V}$$



$$v_1 = 20 - 10 = 10 \text{ V}$$

power-absorbed for following elements:

(i) 10 A current source : $v_1 \times 10 = 10 \times 10 = 100 \text{ W}$

(ii) 10 V voltage source : $10 \text{ V} \times 10 \text{ A} = 100 \text{ W}$

(iii) v_g voltage source : $(-18) \times (-6 \text{ A}) = 108 \text{ W}$

(iv) 6 A current source : $2 \text{ V} \times 6 \text{ A} = 12 \text{ W}$

(v) $10v_a$ dependent source : $20 \text{ V} \times (-16 \text{ A}) = -320 \text{ W}$

1-mark for each correct value of power-absorbed

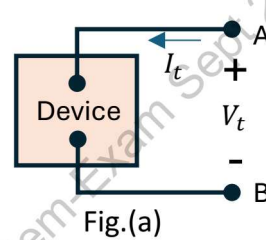
No-partial grading.

Deduct full marks even if sign is incorrect.

Q3: The voltage and current were measured at the terminals of the device shown in Fig.(a). The results are tabulated in Fig.(b).

Obtain the Norton's equivalent model of this device. **(5-marks)**

[**Caution:** Notice the direction of current]



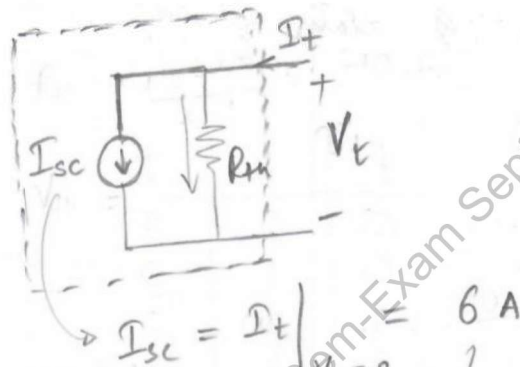
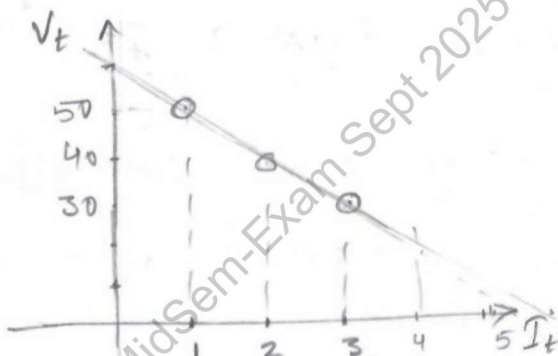
V_t (V)	I_t (A)
50	1
40	2
30	3

Fig.(a)

Fig.(b)

Notice that : as the current injected into the device increases, the terminal voltage drops !!

Graphical Approach



1-mark for correct value of I_{sc} and 1-mark for correct polarity.

can be obtained using extrapolating the V_t - I_t curve.

1-mark for correct approach.

Now, we can use one of the measured values to find R_{th} :

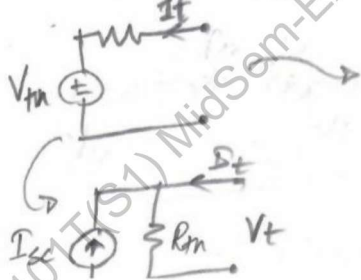
$$V_t = R_{th} \times (I_t - I_{sc})$$

$$\therefore 50 = R_{th} \times (1 - 6) \Rightarrow \underline{R_{th} = -10 \Omega}$$

2-marks for correct value of R_{th} . No partial marks.

OR The device terminal quantities are given as per passive sign convention.

$$R_{th} = \frac{V_2 - V_1}{I_2 - I_1} = \frac{40 - 50}{2 - 1} = -10 \Omega$$



$$\text{Here, } V_t = V_{th} + R_{th} I_t \Rightarrow V_{th} = V_t - R_{th} I_t = 50 - (-10)(1) = 60 \text{ V}$$

$$\therefore I_{sc} = \frac{V_{th}}{R_{th}} = \frac{60}{-10} = -6 \text{ A}$$

Q4: A digital lock opens only when both the correct password switch (P) is ON and the master key switch (K) is ON. For all other cases, the lock remains closed. **(1+2+2+3 marks)**

- Define and name Boolean variables for both the inputs and output.
- Construct the truth table and write the Boolean expression for the lock output (closed/open).
- Realize this logic using only NAND gates.
- Implement the obtained logic using MOSFET based circuit.

(a) Let password switch $\rightarrow P$ which can be ON or OFF
 $\rightarrow$ "1" or "0"
 master key switch $\rightarrow K$ which can be ON or OFF
 $\rightarrow$ "1" or "0"
 status of lock $\rightarrow Z$ which can be open or closed
 $\rightarrow$ "1" or "0"

deduct 0.5 mark for each missing definition

(b)

P	K	Z
0	0	0
0	1	0
1	0	0
1	1	1 (OPEN)

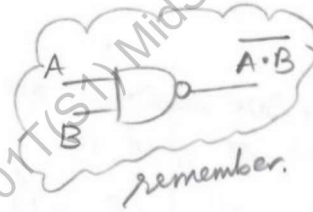
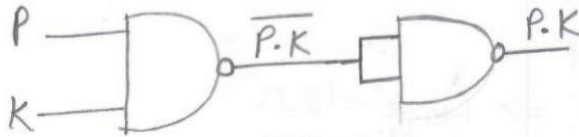
closed

$$Z = P \cdot K \quad (\text{using SOP})$$

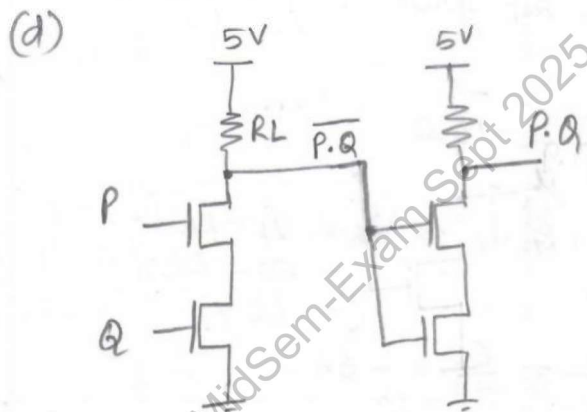
0.5 mark for each correct entry of Z (Output variable)

(c)

$$Z = P \cdot K = \overline{\overline{P \cdot K}}$$



deduct 1-mark for each incorrect notation.



deduct 1-mark for each incorrect notation/ connection.

If only NOT gate is used in second stage, deduct 1-mark

Q5: An inverter circuit (implementation of NOT-Gate) is shown in Fig.(a) and its $v_O - v_I$ relationship is shown in Fig.(b). **(1+1+2+3 marks)**

- Determine the threshold voltage of the MOSFET.
- Which model of MOSFET is used here (S, SR, OR SCS?)
- Determine the value of ON-state resistance of the MOSFET.
- State with justification if this NOT-Gate satisfies the following static discipline:

$$V_{IL} = 2 \text{ V}, V_{IH} = 4 \text{ V}, V_{OL} = 0.2 \text{ V}, V_{OH} = 4 \text{ V}$$

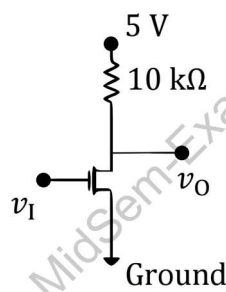


Fig.(a)

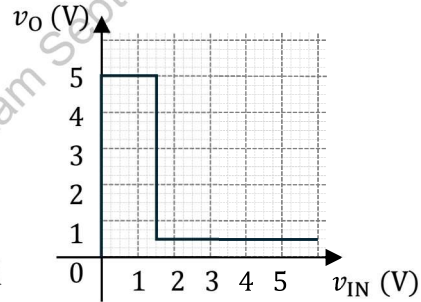


Fig.(b)

- (a) using fig(b), we see that V_O transits sharply at 1.5 V,
 $\therefore$ threshold voltage $V_T = 1.5 \text{ V}$

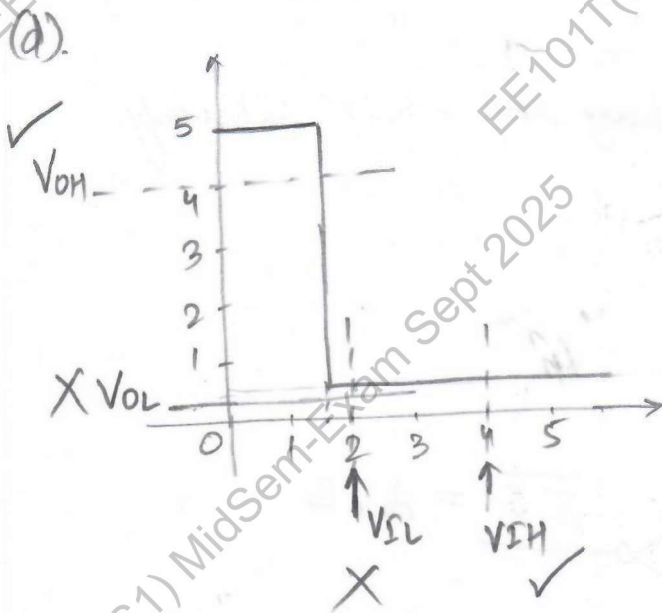
1-mark for correct value. accept 1.4 - 1.6 V as the answer.

- (b) Since the on-state voltage across D & S terminals is not-zero, SR-model of MOSFET is used.

1-mark for correct response.

(c) $V_{DS,ON} = 0.5 \text{ V} = \frac{R_{ON}}{R_{ON} + R_L} \cdot 5$
 $\therefore 0.1 = \frac{R_{ON}}{R_{ON} + 10,000} \Rightarrow R_{ON} = 1.11 \text{ k}\Omega$

2-mark for correct answer. No partial grading.



- (i) $V_{IL} = 2 \text{ V}$: No
 2V is detected as high, not Low, hence, not satisfying.

- (ii) $V_{IH} = 4 \text{ V}$: Yes
 voltages above 4V are detected as high, hence satisfying.

- (iii) $V_{OL} = 0.2 \text{ V}$, No
 During output Low, the value is 0.25 which is more than 0.2V, hence not satisfying.

- (iv) $V_{OH} = 4 \text{ V}$, Yes
 During output high, the value is 5V > 4V. hence, satisfying.

Deduct 1-mark for each incorrect response.

Do not evaluate the response which has no justification given.

Q6: A truth-table showing relationship between output Z and inputs A , B , and C is given here. **(2+2+2-marks)**

- Derive the expression for the logic of Z .
- Minimize the obtained expression such that it can be realized using AND and NOT Gates only.
- Realize the minimized expression using AND and NOT Gates only.

A	B	C	$Z = F(A, B, C)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

(a) Using SOP (Sum-of-products) Approach, we get

$$Z = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$$

2-marks for correct response. Deduct 0.5 marks for each incorrect term. (there are total 6 terms)

(b) Minimization of logic

$$Z = \bar{A}B(\bar{C}+C) + A\bar{B}(\bar{C}+C) + AB(\bar{C}+C)$$

$$= \bar{A}B + A\bar{B} + AB$$

$$= \bar{A}B + A(\bar{B}+B)$$

$$\bar{A}B + A$$

$$= A + B$$

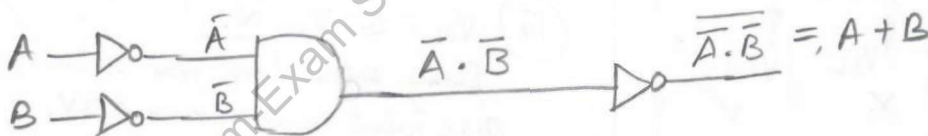
Give only 1-mark if logic is minimized till this point.

Give full 2-marks if the logic is minimized to this point

Now, this needs to be realized using AND & NOT Gates only.

$$Z = A + B = \overline{\overline{A+B}} = \overline{\bar{A} \cdot \bar{B}}$$

(c)

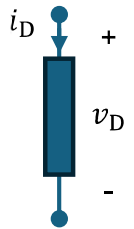


Give 2-marks for correct response. This response should be in accordance with the minimized logic obtained in part (b). Deduct 1-mark for each incorrect notation/connection

Q7: The terminal-characteristic ($v_D - i_D$ relationship) of an element shown in Fig. (a) is given as follows: **(1+2+3-marks)**

$$v_D = k i_D^2$$

- Determine the SI-unit of k .
- Given that $k = 50$, derive the small-signal model of this element around the operating current of $I_D = 0.1$ A.
- Draw the small-signal equivalent model of this element and label the key parameters with their SI-units.



(a) $[k] = \left[\frac{V}{A^2} \right]$, volt per ampere squared.

1-mark for correct response.

(b) $k = 50 \frac{V}{A^2}$, $I_D = 0.1$ A

$i_D = I_D + \Delta i_D$ & corresponding $v_D = V_D + \Delta v_D$

$V_D = k I_D^2 = 50 \times (0.1)^2 = 0.5$ V

1-mark for procedure

$$v_D \approx v_D \Big|_{i_D=I_D} + \frac{dv_D}{di_D} \Big|_{i_D=I_D} \cdot \Delta i_D = 0.5 + k i_D \Big|_{i_D=I_D} \cdot \Delta i_D$$

$$V_D + \Delta v_D = 0.5 + 100 \times 0.1 \cdot \Delta i_D$$

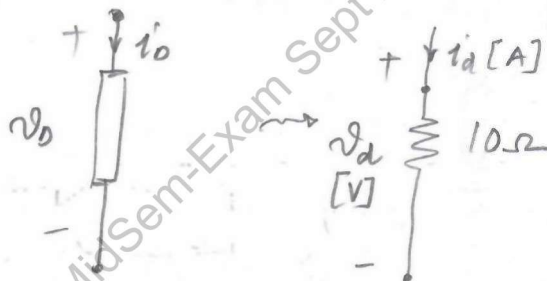
$\therefore \Delta v_D = 10 \Delta i_D$

We can rename Δv_D as v_d & Δi_D as i_d
and we get then $v_d = 10 i_d$

1-mark for correct relationship

(c)

The part(c) is to be corrected only if part(b) is correct.



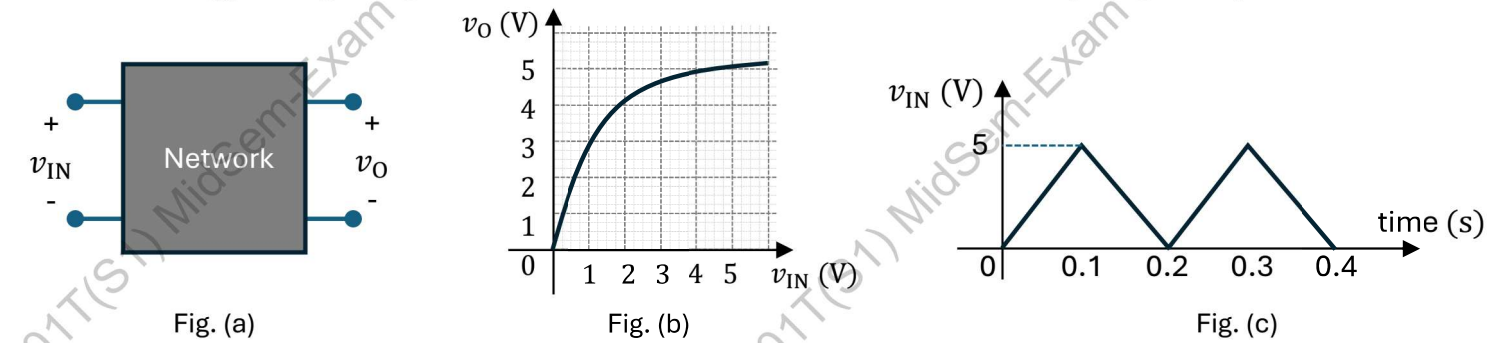
1-mark for correct unit of resistor.

1+1 mark for correct notation and polarity of the voltage and current. This is to be graded only if resistor unit is correct.

actual Element
[non-linear]

Small-signal equivalent
[linear]

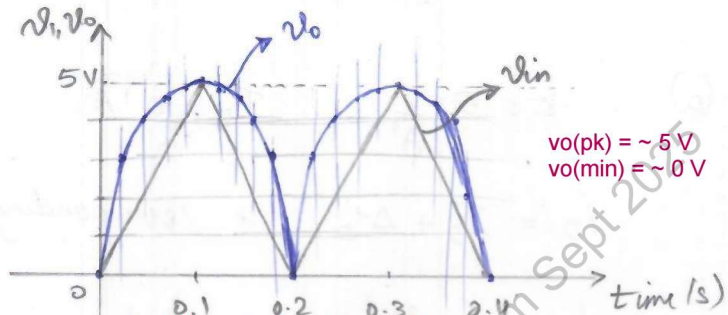
Q8: Transfer characteristic of a network shown in Fig.(a) is given in Fig.(b). A triangular-waveform with the peak value of 5 V, as shown in Fig.(c), is applied across the input terminals of this network. Sketch the waveforms of v_{IN} and v_O , with the same horizontal-axis. Label the sketch neatly. **(5-marks)**



Both, the input and output should be shown with common x-axis (time-axis)

Approach 1.

time	$v_{in}[V]$	$v_o[V]$
0	0	0
0.02	1	~ 3
0.04	2	~ 4
0.06	3	~ 4.6
0.08	4	~ 4.9
0.10	5	~ 5.1

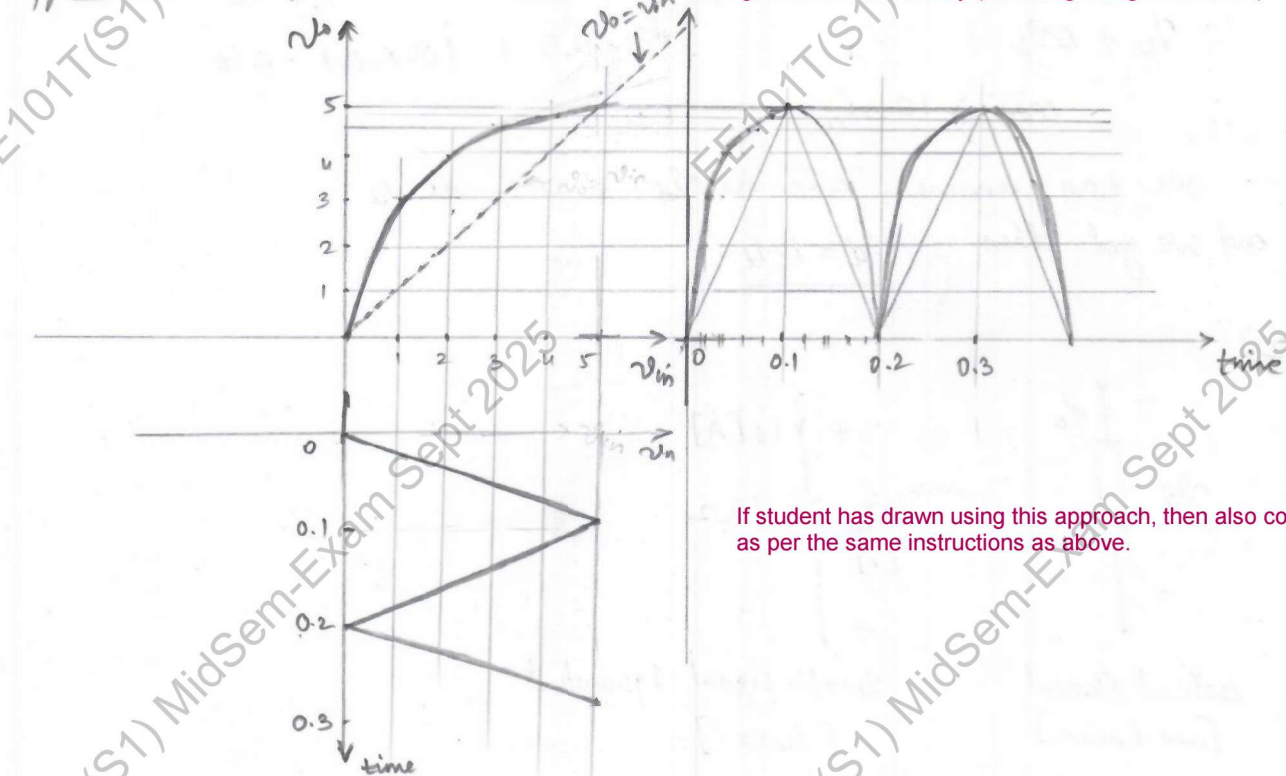


Award directly 0 marks if the output waveform is triangular.

If output looks flat-topped, as shown above, then evaluate as per the following.

- Deduct 2-marks if input voltage waveform is not shown along with output voltage.
- Deduct 1-mark if the curves (v_i and v_o) are not labelled.
- Deduct 1-mark if $v_o(pk)$ is less than 5V
- Deduct 1-mark if the figure is drawn casually (not using straight-lines, etc.)

Approach 2

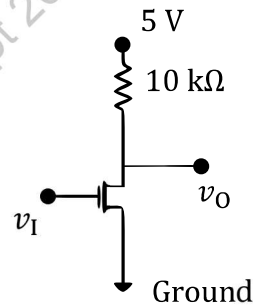


If student has drawn using this approach, then also correct it as per the same instructions as above.

Q9: The following circuit is to be used for amplifying input signal v_i . The parameters associated with the MOSFET used here are as follows: **(3+2 and 6-Bonus marks)**

Material related parameter: $K = 0.5 \frac{\text{mA}}{\text{V}^2}$, and Threshold voltage: $V_T = 2 \text{ V}$

- Determine the valid range (minimum and maximum) of input voltage that ensures that the MOSFET is in its saturation region.
- Determine the corresponding output voltage range (minimum and maximum).
- Sketch the output versus input curve (v_o vs. v_i) for this circuit considering the input range of $-5 \leq v_i \leq 5$. [**Bonus problem**]



We know that to use MOSFET in an amplifier circuit, we have to operate it in saturation region, i.e. $V_{GS} \geq V_T$ & $V_{DS} \geq (V_{GS} - V_T)$

(a) $V_{i,\min} \leq V_{GS} \leq V_{i,\max}$, here $V_{GS} = V_i$

$V_{i,\min} = V_T$ (threshold voltage). = 2V

1-mark for correct value of minimum

$V_{i,\max}$ occurs when $v_o = v_i - V_T$

for the circuit shown, we have, $v_o = 5 - R_D \cdot i_D$

in saturation region $i_D = \frac{K}{2} \cdot (V_{GS} - V_T)^2 = \frac{K}{2} (v_i - V_T)^2$

1-mark for procedure

We have $\frac{V_{i,\max} - V_T}{2} = -\frac{K}{2} (V_{i,\max} - V_T)^2 \cdot R_D + 5$

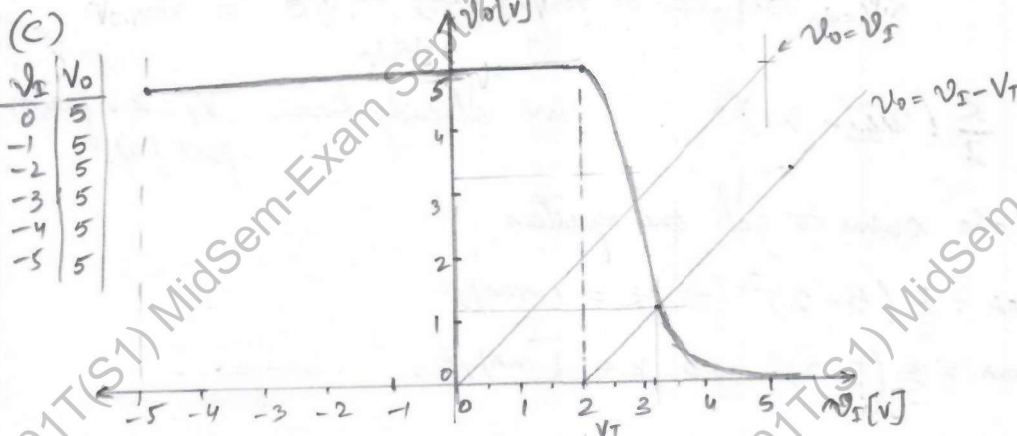
$\therefore x = -\frac{0.5 \times 10^{-3} \cdot 10,000}{2} x^2 + 5$

$2.5x^2 + x - 5 = 0 \rightarrow x = 1.228 \rightarrow V_{i,\max} = \underline{3.228 \text{ V}}$

1-marks for correct value of maximum

(b) $V_{o,\min} = V_{i,\max} - V_T = 1.228 \text{ V}$, $V_{o,\max} = 5 \text{ V}$ (for $V_{in} = V_T$)

1-mark each for correct values of corresponding output minimum and maximum



Deduct 1-mark if V_T is not denoted.

Deduct 1-mark if labels are missing.

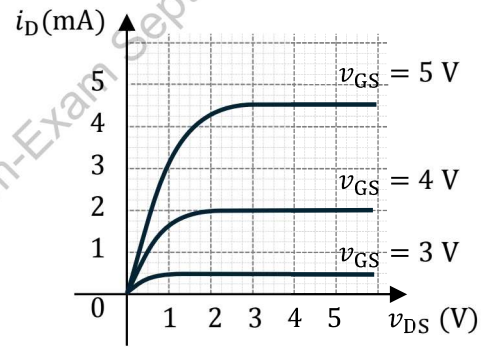
Deduct 1-mark if the transition from Saturation to triode region is not shown and doesn't match with previous answers

Deduct 2-marks if the characteristics is not shown for negative value of input

Q10: The $i_D - v_{DS}$ curves of a MOSFET are captured using experiments for different values of v_{GS} . The results are shown in the figure, where the corresponding values of v_{GS} are also marked.

Using this results: **(4+1+3-marks)**

- Find the value of v_{GS} below which MOSFET enters the cut-off region.
- For $v_{DS} = 2$ V, find the value of v_{GS} above which MOSFET enters the triode-region.
- Determine the values of K and V_T in the expression of current $i_D = \frac{K}{2}(v_{GS} - V_T)^2$. Recall that this expression is valid only in the saturation region of MOSFET.



(a) MOSFET enters the cut-off region when $V_{GS} < V_T$, V_T : Threshold voltage
 therefore here, we need to find out V_T .
 we can use information from the saturation region of the shown graph
 in this region $i_D = \frac{K}{2}(V_{GS} - V_T)^2$

1-mark for recognizing that V_T is to be calculated

we can get two such equations using curves shown as follows.

$$4.5 \text{ mA} = \frac{K}{2}(5 - V_T)^2 \quad \text{--- ①}$$

2-marks for correct approach

$$2 \text{ mA} = \frac{K}{2}(4 - V_T)^2 \quad \text{--- ②}$$

using: ① & ② $\frac{4.5}{2} = \left(\frac{5 - V_T}{4 - V_T}\right)^2 \rightarrow 1.5 = \frac{5 - V_T}{4 - V_T} \rightarrow V_T = 2 \text{ V}$

1-mark for correct answer of V_T

(b) $v_{DS} = 2 \text{ V}$ | MOSFET enters triode-region when $V_{GS} \geq V_{GS} - V_T$

The boundary is at $V_{DS} = V_{GS} - V_T$.

Since $V_T = 2 \text{ V}$ & $V_{DS} = 2 \text{ V} \Rightarrow V_{GS} = 4 \text{ V}$

1-mark for correct value

This can be verified using the plot as well.

(c) $i_D = \frac{K}{2}(V_{GS} - V_T)^2$, we already know $V_T = 2 \text{ V}$ from part (a).

use any of the curve to get one equation

$$4.5 \text{ mA} = \frac{K}{2}(5 - 2)^2 \Rightarrow K = 1 \text{ mA/V}^2$$

1-mark for correct value of V_T .

OR $0.5 \text{ mA} = \frac{K}{2}(3 - 2)^2 \Rightarrow K = 1 \text{ mA/V}^2$

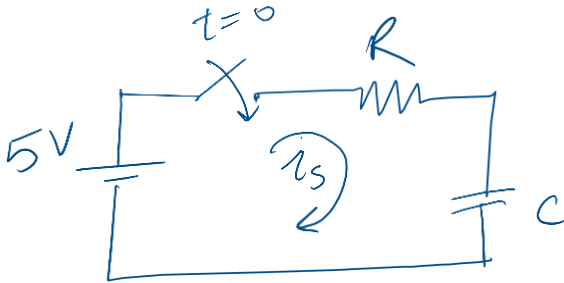
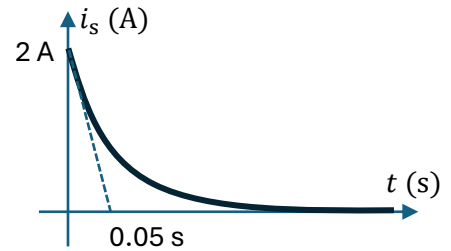
1-mark for correct value of K and 1-mark for correct unit.

No partial grading here.

Name: _____, Roll No: _____, Total marks: 30, Duration: 35 min

Note: Make and state suitable assumptions in case of any missing information.

Q1: An *RC*-series circuit is connected to a constant DC-source at $t = 0$ s. The magnitude of this source is 5 V. The current drawn from the source is shown in the figure. Determine the values of R and C of this *RC*-circuit. **(3-marks)**



at $t=0$, $i_s = \frac{5}{R} \Rightarrow R = \frac{5}{2} \Omega$
 1-mark for R

from the figure, we see that
 $\tau = 0.05 \text{ s}$, 1-mark for recognizing tau

for *RC*-series circuit, $\tau = RC$
 $\therefore C = \frac{\tau}{R} = \frac{0.05}{2.5} = 0.02 \text{ F}$

1-mark for C

Name: _____, Roll No: _____, Total marks: 30, Duration: 35 min

Note: Make and state suitable assumptions in case of any missing information.

Sree Vani

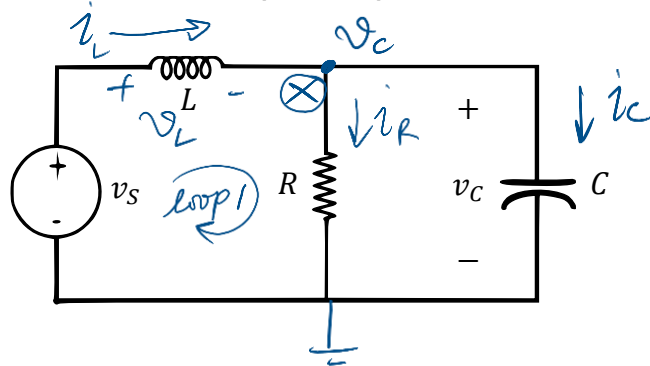
Q2: For the circuits shown below:

(a) Obtain the differential-equation representing the system dynamics in terms of v_C . **(4-marks)**

(b) Obtain the characteristic-equation in the terms of complex-frequency s . **(2-marks)**

(c) Determine with justification the nature of the response* for $R = 10 \Omega$, $L = 100 \text{ mH}$, and $C = 10 \mu\text{F}$.

*(under-damped, critically-damped, over-damped) **(2-marks)**



a)

① $v_s - L \frac{di_L}{dt} - v_C = 0$ (KVL in first loop) 1.5-mark for correct KVL/KC

② $i_L = i_R + i_C$ (KCL at node X)
 $= \frac{v_C}{R} + C \frac{dv_C}{dt}$ 1.5-mark for correct KVL/KC

$L \frac{di_L}{dt} + v_C = v_s$

$L \frac{d}{dt} \left(\frac{v_C}{R} + C \frac{dv_C}{dt} \right) + v_C = v_s$ 1-mark for the complete DE

OR $\frac{d^2 v_C}{dt^2} + \frac{1}{RC} \frac{dv_C}{dt} + \frac{v_C}{LC} = \frac{v_s}{LC}$ — (E1)

b) Let's assume $v_C = A e^{st}$ is the solution to Equation (E1)

$\frac{d^2 v_C}{dt^2} + \frac{1}{RC} \frac{dv_C}{dt} + \frac{v_C}{LC} = 0$

1-mark for procedure

$\therefore A s^2 e^{st} + \frac{1}{RC} A s e^{st} + \frac{A e^{st}}{LC} = 0 \Rightarrow \left(s^2 + \frac{s}{RC} + \frac{1}{LC} \right) = 0$ --- (E2)

1-mark for the above equation

c) $R = 10, L = 100 \text{ mH}, C = 10 \mu\text{F}$

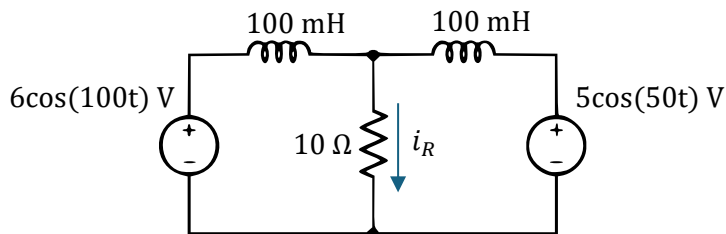
Comparing (E2) with standard form $s^2 + 2\alpha s + \omega_0^2 = 0$ gives, $\alpha = \frac{1}{2RC}$ & $\omega_0 = \frac{1}{\sqrt{LC}}$ 1-mark for procedure

$\alpha = \frac{1}{2 \times 10 \times 10 \times 10^{-6}} = 5000 \text{ rad/s}$, & $\omega_0 = \frac{1}{\sqrt{100 \times 10^{-3} \times 10 \times 10^{-6}}} = 1000 \text{ rad/s}$

Since $\alpha > \omega_0$, response is over-damped. 1-mark for correct response.

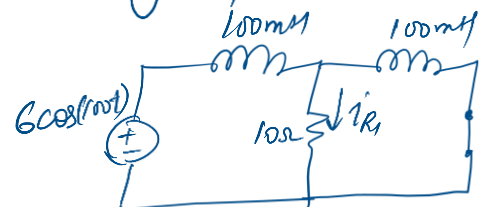
Q3: In the given circuit, determine the expression of current through resistor, i_R during the steady-state.
(6 marks)

Vijay

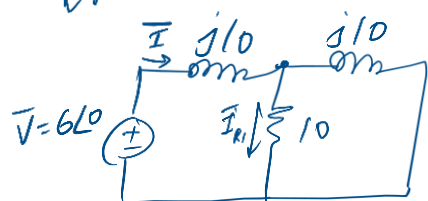


using superposition

1-mark if students have recognized to use superposition principle and drew two sub.



phasor, $\omega = 100$



$$\bar{I}_{R1} = \bar{I} \times \left(\frac{j10}{j10 + 10} \right) = \frac{6\angle 0}{j10 + \left(\frac{j10 \times 10}{j10 + 10} \right)} \left(\frac{j(10)}{j10 + 10} \right)$$

1-mark for procedure
 $= 0.268 \angle -26.56^\circ \text{ A}$

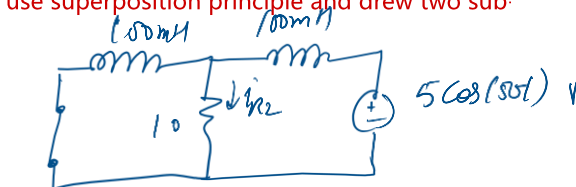
$$i_{R1} = 0.268 \cos(100t - 26.56^\circ)$$

1-mark for time-domain expression of the current

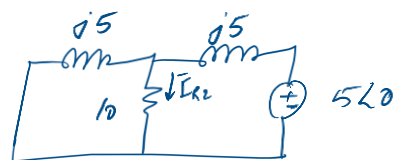
$$i_R = i_{R1} + i_{R2} = 0.268 \cos(100t - 26.56^\circ) + 0.243 \cos(50t - 14.03^\circ) \text{ A}$$

1-mark for adding the time-domain expressions.

If student has added phasors directly, then deduct 3 marks



phasor $\omega = 50$



$$\bar{I}_{R2} = \left(\frac{j5}{10 + j5} \right) \cdot \bar{I} = \frac{j5}{10 + j5} \times \frac{5\angle 0}{j5 + \left(\frac{10 \times j5}{10 + j5} \right)}$$

1-mark for procedure
 $= 0.243 \angle -14.03^\circ \text{ A}$

$$i_{R2} = 0.243 \cos(50t - 14.03^\circ)$$

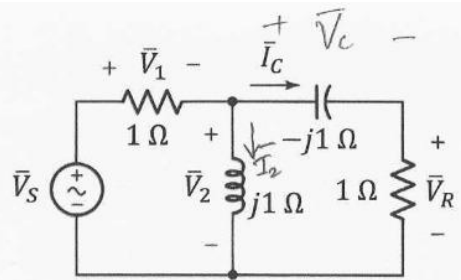
1-mark for time-domain expression of the current

Noor

Q4: Assuming $\bar{I}_C = 2\angle 0^\circ$ A (rms), draw a phasor diagram

(to the scale) showing $\bar{I}_C$, $\bar{V}_R$, $\bar{V}_2$, $\bar{V}_1$, and $\bar{V}_S$. (4 marks)

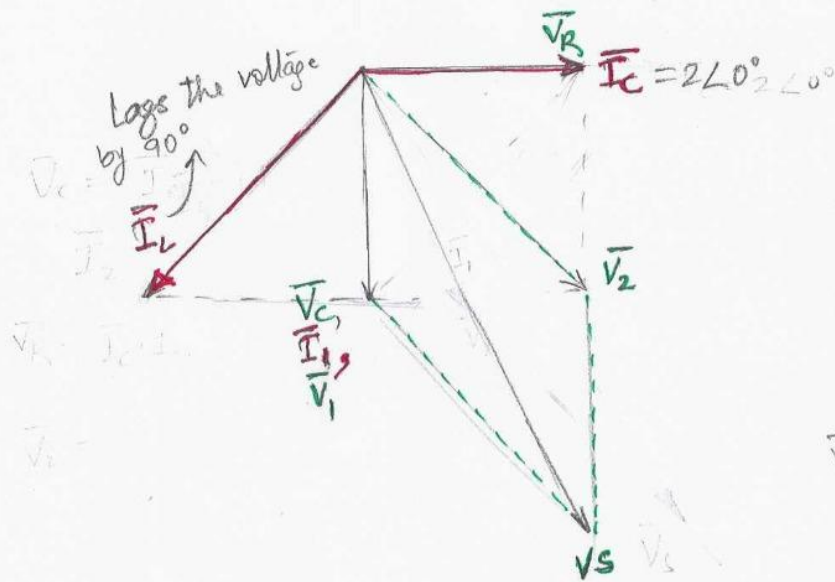
Measure and write the ratio of the lengths (using scale) of
(a) $\bar{V}_S$ to $\bar{V}_1$ (b) $\bar{V}_1$ to $\bar{V}_2$; (c) $\bar{V}_S$ to $\bar{V}_R$. (3 marks)



$$\bar{V}_C = (-j1) \cdot \bar{I}_C$$

2-marks for procedure

$$\bar{V}_R = \bar{I}_C \cdot 1$$



0.5 marks for each correct phasor

$$\bar{V}_2 = \bar{V}_R + \bar{V}_C$$

$$\bar{I}_2 = \frac{\bar{V}_2}{j}$$

$$\bar{V}_S = \bar{V}_1 + \bar{V}_2$$

$$\left| \frac{\bar{V}_S}{\bar{V}_1} \right| = \frac{78 \text{ mm}}{35 \text{ mm}} = 2.23$$

1-mark

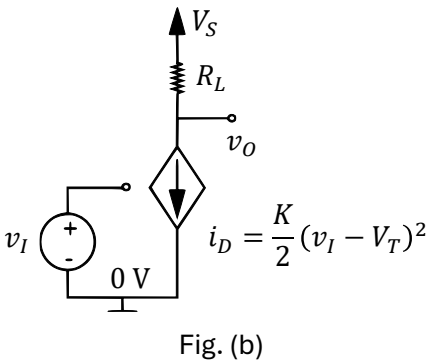
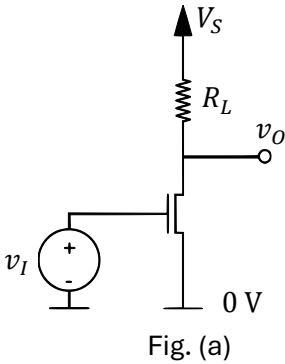
$$\left| \frac{\bar{V}_1}{\bar{V}_2} \right| = \frac{35 \text{ mm}}{49 \text{ mm}} = 0.714$$

1-mark

$$\left| \frac{\bar{V}_S}{\bar{V}_R} \right| = \frac{78 \text{ mm}}{34 \text{ mm}} = 2.29$$

1-mark

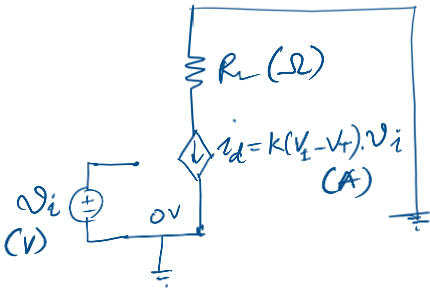
Q5: The MOSFET amplifier and its large-signal model (SCS) are shown here in Figs. (a) and (b), respectively. Draw the small-signal equivalent model of this amplifier and label all the elements and the associated quantities along with their units. Mention the necessary assumptions, if any. **(6 marks)**



1-mark for the units of v_i , R_L and i_d

2-marks for the circuit

1-mark for approach



operating point is $v_I = V_I$

1-mark if the supply voltage is replaced by

1-mark for i_d expression

Indian Institute of Technology Dharwad

End-Semester Examination – Nov 2025

EE101T(S1): Introduction to Electrical Systems and Electronics

(Instructor: Amarkumar Kushwaha)

Full Name (All caps): Solution, Roll No: —

24th Nov 2025, 9:30 AM – 12:30 PM

Total marks: 70

Duration: 180 minutes

Instructions:

- There are total **12** questions on **3-printed sides**.
- Begin a **new-question from new-page**, clearly mentioning the question number.
- If any information is missing, make **reasonable assumptions** and state the same with justification.
- Except for scientific calculators, **NO Electronics gadgets** are allowed.
- Mention the **unit** of all the quantities in the intermediate-steps and the final-answers.

0.5 mark for each correct response

Maximum marks is 5

Q1(a): Fill the third column of the following table selecting the correct option from fourth column. (3-marks)

Sr. No.	Electrical device	Scale of power consumption	Options
1	Household Ceiling Fan	$\sim 20-50W$	$\sim 1 \mu W$
2	Electric iron	$\sim 1kW$	$\sim 10 mW$
3	Generators in thermal power-plants	$\sim 100 MW$	$\sim 5 W$
4	Scientific calculator	$\sim 1 \mu W$	$\sim 20-50 W$
5	Indian Railways Locomotive	$\sim 5 MW$	$\sim 300 W$
6	A single LED	$\sim 10 mW$	$\sim 1 kW$
7	A smart-watch	$\sim 10 mW$	$\sim 5-10 kW$
8	Electric bicycle motor	$\sim 300W$	$\sim 100 kW$
			$\sim 5-8 MW$
			$\sim 100 MW$

Q1(b): Determine the **UNIT** of the following quantities, giving 1-line justification. (5-marks)

Sr. No.	item/function
1	Voltage phasor $\bar{V}$ and current phasor $\bar{I}$
2	Ratio of v_R and i_R , where v_R is the voltage across a resistor and i_R is current through it
3	Ratio of v_L and i_L , where v_L is the voltage across an inductor and i_L is current through it
4	Ratio of v_C and i_C , where v_C is the voltage across a capacitor and i_C is current through it
5	Ratio of $\bar{V}_L$ and $\bar{I}_L$, where $\bar{V}_L$ and $\bar{I}_L$ are the voltage and current phasors associated with an inductor

1-mark for each correct response. Give 0 marks if no justifications is given.

(1) $\bar{V} = V e^{j\theta} \rightarrow V_0, \text{ volt}, \quad \bar{I} = I e^{j\theta} \rightarrow A, \text{ ampere}$

(2) $\frac{v_R}{i_R} = \text{Resistance} \rightarrow \frac{V}{A}$

(3) $v_L = L \frac{di_L}{dt} \rightarrow V \text{ \& } i_L \rightarrow A : \left[\frac{v_L}{i_L} \right] \rightarrow \text{unit of } \frac{V}{A}$

(4) $i_C = C \frac{dv_C}{dt} \rightarrow A \text{ \& } v_C \rightarrow V : \left[\frac{v_C}{i_C} \right] \rightarrow \text{unit of } \frac{V}{A}$

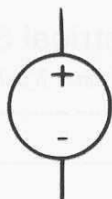
(5) $\frac{\bar{V}_L}{\bar{I}_L} = jX_L \rightarrow \text{unit of } \frac{V}{A} \text{ or ohm, } \Omega$

No partial marks here.

Q2: Draw the small-signal equivalent models of the following elements. Clearly denote the terminal quantities and element values with their unit. **(4-marks)**



(a)



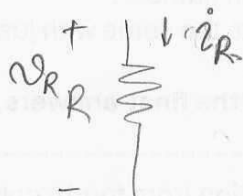
(b)



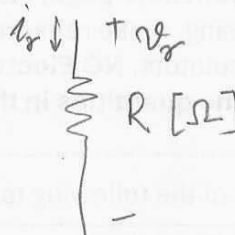
(c)

for all of the following, deduct 0.5 marks if terminal quantities Or units are missing.

(a)

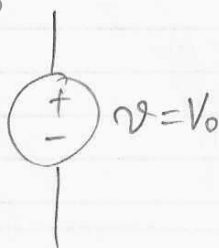


SS model.

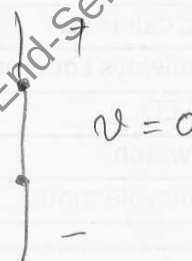


1-mark for correct response

(b)



SS model



1-mark for correct response

(c)



SS model



2-marks for correct response

No partial marks here.

$$v_D = 20 - 2.5 i_D^2$$

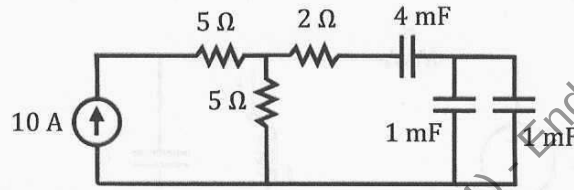
$$v_D = V_D + v_d$$

$$i_D = I_D + i_d$$

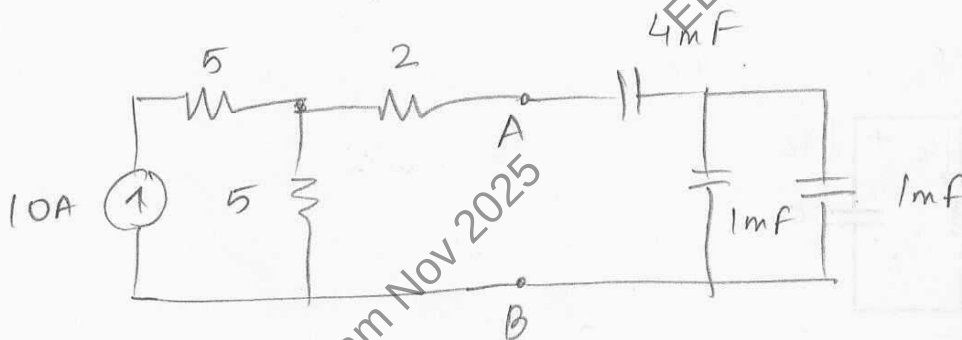
$$v_D = (20 - 2.5 i_D^2) + (-5 I_D) \cdot i_d$$

$$v_D = V_D + (-5 I_D) \cdot i_d$$

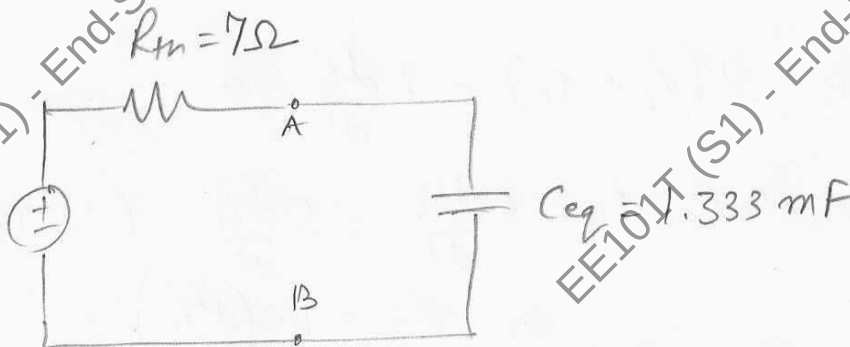
Q3: Determine the value of time-constant for the following circuit. (4-marks)



For calculation of time constant, we require R_{th} & C_{eq} .



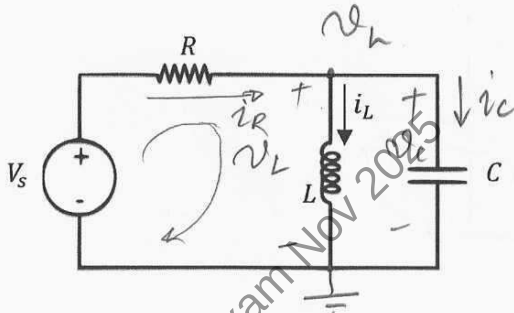
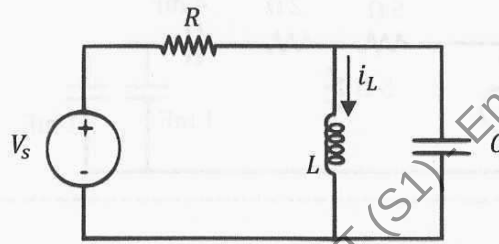
2-marks for the procedure if the final answer is incorrect.



$$\therefore \text{time constant, } \tau = R_{th} \cdot C_{eq} = 7 \times 1.33 \text{ ms} = 9.33 \text{ ms}$$

4-marks directly if the answer is correct.
Deduct 1-mark if the unit of time-constant is missing.

Q4: For the circuit shown below, obtain the differential-equation representing the system dynamics in terms of i_L . Determine the expression for resonant-frequency and neper-frequency. (4-marks)



KVL in loop $\rightarrow V_s - R i_R - v_L = 0$

1-mark for correct KVL/KCL

$$\therefore V_s - R(i_L + i_C) - L \frac{di_L}{dt} = 0 \quad \text{--- ①}$$

We know that $i_C = C \frac{dv_C}{dt} = C \frac{dv_L}{dt} \quad (\because v_C = v_L)$

Substitution gives

$$\rightarrow V_s - R i_L - R L C \frac{d^2 i_L}{dt^2} - L \frac{di_L}{dt} = 0$$

1-mark for another KVL/KCL

$$\frac{d^2 i_L}{dt^2} + \frac{1}{RC} \frac{di_L}{dt} + \frac{1}{LC} i_L = \frac{V_s}{RLC}$$

1-mark for correct final response

Comparing the above Equation with standard form.

$$\frac{d^2 x}{dt^2} + 2\alpha \frac{dx}{dt} + \omega_0^2 x = \text{gives}$$

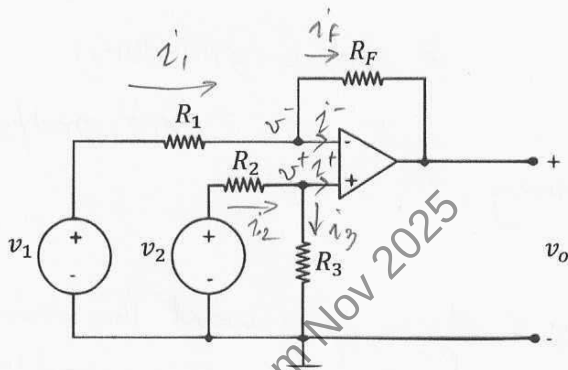
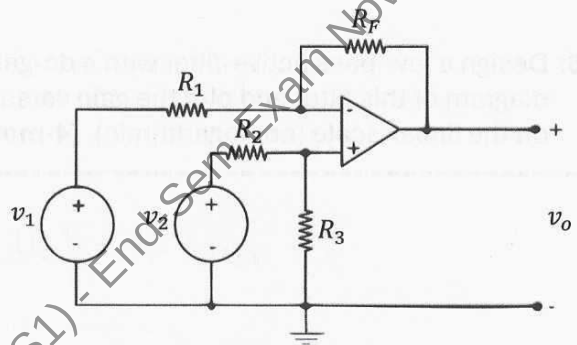
neper freq: $\alpha = \frac{1}{2RC} \text{ [rad/s]}$

0.5 mark each for the expressions of frequencies.

resonant freq: $\omega_0 = \frac{1}{\sqrt{LC}} \text{ [rad/s]}$

Deduct 0.5 marks if units of any of the frequencies are missing.

Q5: For the circuit shown here, derive the expression for v_o in terms of inputs v_1 , v_2 , R_1 , R_2 , R_3 , and R_F . Assume that the Op-Amp is ideal. (4-marks)



$$i^- = 0, i^+ = 0, v^- = v^+$$

$$\therefore i_1 = i_F$$

$$\therefore \frac{v_1 - v^-}{R_1} = \frac{v^- - v_o}{R_F}$$

$$\therefore \frac{v_1 - v^+}{R_1} = \frac{v^+ - v_o}{R_F} \quad \text{--- (1)}$$

1-mark for obtaining one relations

Also, $i_2 = i_3$ or we can use voltage-divider rule at v^+ node.

$$v^+ = \frac{R_3}{R_2 + R_3} \cdot v_2 \quad \text{--- (2)}$$

1-mark for obtaining another relationship

Putting (2) into (1) gives the following.

Let's first simplify (1) first, then substitute (2)

$$\frac{v_1 - v^+}{R_1} = \frac{v^+ - v_o}{R_F} \quad \therefore \frac{v_o}{R_F} = -\frac{v_1}{R_1} + \left(\frac{1}{R_1} + \frac{1}{R_F}\right)v^+$$

$$\therefore \frac{v_o}{R_F} = -\frac{v_1}{R_1} + \left(\frac{1}{R_1} + \frac{1}{R_F}\right) \cdot \left(\frac{R_3}{R_2 + R_3}\right) \cdot v_2$$

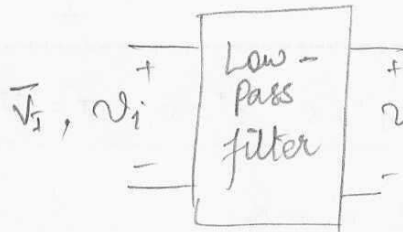
1-mark for procedure

$$v_o = R_F \left[\left(\frac{R_1 + R_F}{R_1 \cdot R_F} \cdot \frac{R_3}{R_2 + R_3} \right) \cdot v_2 - \frac{v_1}{R_1} \right]$$

1-mark for correct final expression

Q6: Design a low-pass active filter with a dc-gain of 8 and the corner frequency of 500 Hz. Draw the circuit diagram of this filter and plot the gain versus frequency plot of this filter in the range of 10 Hz to 2000 Hz on the linear-scale (not logarithmic). **(4-marks)**

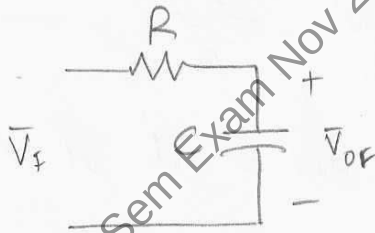
We need the following.



$$\frac{|V_o|}{|V_i|} \approx 8 \text{ at lower frequency } < 500 \text{ Hz.}$$

$$\omega_c = 2\pi \cdot (500 \text{ Hz}) = 3,142 \text{ rad/s.}$$

We can realize a low-pass filter using RC circuit



here $\omega_c = \frac{1}{RC}$, we want this value to be 3,142 rad/s.

$$\therefore RC = 318 \text{ } \mu\text{s.}$$

Lets choose $R = 1000 \Omega$

$$C = 0.32 \text{ } \mu\text{F}$$

1-mark for identifying the value of R and C

Multiple values of R and C are possible here.

Gain of this filter (passive) is:

$$\frac{1}{\sqrt{1 + \omega^2 R^2 C^2}}$$

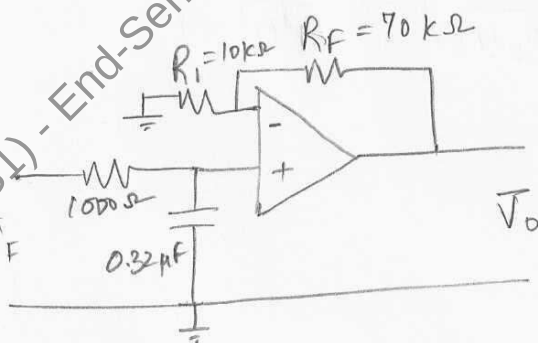
at small frequency, $\omega = 30 \text{ rad/s}$ example,

$$\text{Gain} = \frac{1}{\sqrt{1 + (30)^2 \cdot (1000)^2 \cdot (0.32 \times 10^{-6})^2}} = 0.99995.$$

To Achieve a gain of 8, we need to amplify this signal using Op-Amp based circuit as follow;

1-mark for identifying the use of Op-Amp

Gain of Op-Amp based circuit should be '8'.



2-marks for correct final circuit

No partial marks here.

multiple solutions are possible here.

Verify that the gain of the amplifier is 8 and the corner frequency of the RC filter is 500 Hz.

Q7(a): A certain load absorbs the complex-power of $\bar{S} = 200 + j100$ VA. The steady-state voltage across this load is $v = 200 \sin(314 \cdot t)$ V. Obtain the peak-value of the current through this load. **(3-marks)**

$$\bar{S} = 200 + j100 \text{ VA}$$

$$\bar{S} = \bar{V} \bar{I}^* \rightarrow (\text{Rms}) \text{ quantities}$$

1-mark for the correct expression of complex power.

$$v = 200 \sin(314 \cdot t) \rightarrow \bar{V} = \frac{200}{\sqrt{2}} \angle 0^\circ \text{ V. (peak)}$$

$$\therefore 200 + j100 = \left(\frac{200}{\sqrt{2}}\right) \bar{I}^*$$

$$\bar{I}^* = 1.414 + j0.707 \text{ A}$$

$$\therefore \bar{I} = 1.58 \angle -26.56^\circ \text{ A}$$

1-mark for the correct procedure.

$$\therefore i = 2.23 \sin(314t - 26.56^\circ) \text{ A}$$

Thus peak-value of current is 2.23 A.

1-mark for the correct final value.

Deduct 0.5 marks if unit of current is missing

Q7(b): The voltage across a load and the current through it are given as $v = 2 \cdot \sin(2\pi \cdot 10 \cdot t)$ V and $i = 1 \cdot \sin(2\pi \cdot 20 \cdot t)$ A. Neatly sketch the waveform of the instantaneous-power absorbed by this load. Also determine the value of average power absorbed by this load over one period of voltage waveform. **(6-marks)**

v, i, p

1-mark for correct labels on y-axis

p is the answer that has to be evaluated

instantaneous-power, $p = v \cdot i$.

2-marks for the waveform that looks like this.

1-mark for correct labels on time-axis

Average-value = 0, (Area under the curve of 'p' sums to zero over a cycle.)

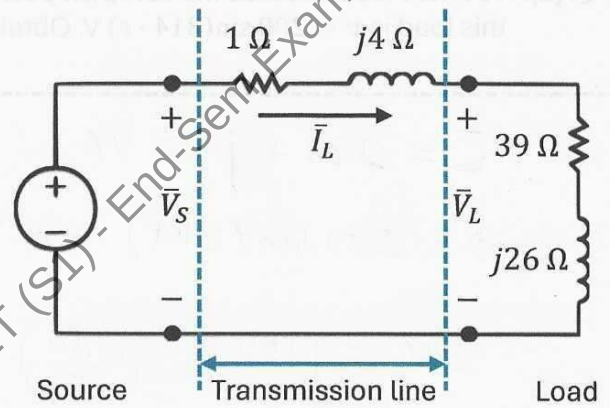
2-marks for the correct average value.

Q8: In the circuit shown here, $\bar{V}_s = 230 \text{ V (rms)}$.

- (a) Determine (i) $\bar{V}_L$, (ii) complex power absorbed by the load $(39 + j26) \Omega$, (iii) active-power dissipated in the load, and (iv) power lost in the transmission line. **(4-marks)**

- (b) Draw the phasor diagram showing $\bar{V}_s$, $\bar{V}_L$ and $\bar{I}_L$ **(3-marks)**

No partial marks here.



$$(a) \quad \bar{I}_L = \frac{\bar{V}_s}{40 + j30} = 3.68 - 2.96j = 4.6 \angle -36.87^\circ \text{ A}$$

$$(i) \quad \bar{V}_L = \bar{V}_s - \bar{I}_L \cdot (1 + j4) = 215.28 - 11.96j = 215.61 \angle -3.2^\circ \text{ V}$$

1-mark for correct value of V_L .

$$(ii) \quad \bar{S}_{\text{Load}} = \bar{V}_L \cdot \bar{I}_L^* = (215.61 \angle -3.2^\circ)(4.6 \angle 36.87^\circ) = 825 + 550j \text{ VA}$$

1-mark for correct value of S_{Load}

$$(iii) \quad \text{active power dissipated in load} = P = 825 \text{ W}$$

1-mark for correct value of P

$$(iv) \quad \text{power lost in transmission line} = |\bar{I}_L|^2 \cdot 1 = 21.2 \text{ W}$$

1-mark for correct value of transmission loss

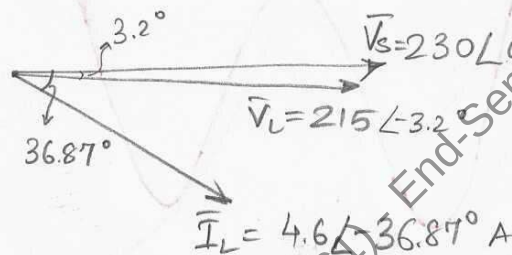
$$\text{or } \bar{S}_{\text{transmission line}} = (\bar{V}_s - \bar{V}_L) \cdot \bar{I}_L^* = 21.2 + 84.64j$$

power lost in transmission line

Deduct 0.5 marks if units of any of the above quantity are missing or incorrect.

(b)

No partial marks here.

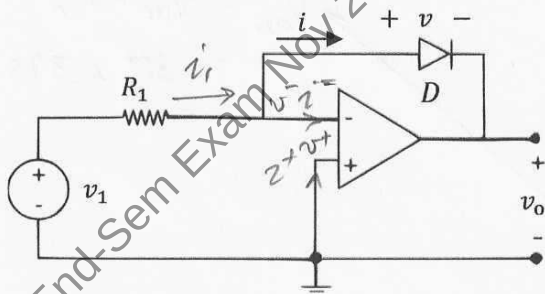
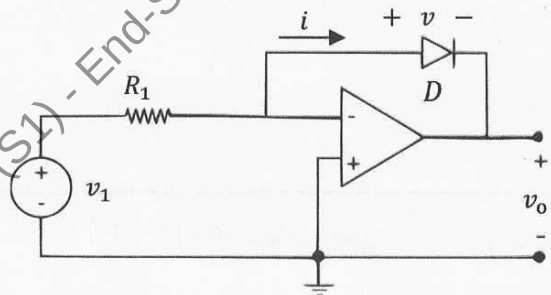


1-mark for each correct phasor, V_s , I_L and V_L

Q9: An Op-Amp based circuit is shown here. The current through the diode ' D ' around the expected operating point can be approximated as $i = I \cdot (e^{kv})$ A, where v is the voltage across the diode, and I and k are some constants. **(5-marks)**

(a) Obtain the expression for v_o in terms of v_1 and other relevant parameters. Assume the Op-Amp to be ideal.

(b) What can you comment on the operation this circuit is performing?



(a)

$$i_1 = i$$

$$\frac{v_1 - v^-}{R_1}$$

$$\text{also } v^- = v^+ = 0,$$

$$\therefore i_1 = R_1 \cdot i \quad \text{--- ①}$$

1-mark for obtaining one expression using KVL/KCL

$$\therefore v_1 = R_1 (I e^{kv})$$

$$\frac{v_1}{R_1 I} = e^{kv} \Rightarrow kv = \ln\left(\frac{v_1}{R_1 I}\right)$$

$$\text{or } v = \frac{1}{k} \ln\left(\frac{v_1}{R_1 I}\right)$$

1-mark for obtaining another expression

$$\text{Now, } v^- - v = 0 - v = -\frac{1}{k} \ln\left(\frac{v_1}{R_1 I}\right)$$

$$v_o = -\frac{1}{k} \ln\left(\frac{v_1}{R_1 I}\right)$$

2-marks for correct final expression

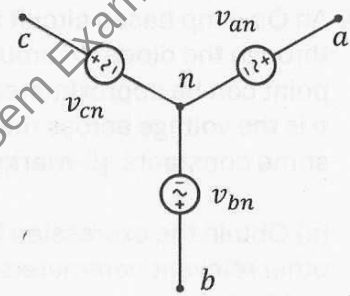
No partial marks here.

(b) The circuit is performing natural logarithm operation on the scaled version of input voltage v_1 .

1-mark for recognizing logarithmic operation.

No partial marks here.

Q10: In the network shown here, $v_{an} = 300 \sin(314 \cdot t)$ V,
 $v_{bn} = 250 \sin(314 \cdot t - 90^\circ)$ V, and $v_{cn} = 200 \sin(314 \cdot t + 120^\circ)$ V.
 Determine the expressions for v_{ab} , v_{bc} , and v_{ca} . **(6-marks)**
 [Hint: Use phasor-diagram based approach]



$v_{an} = 300 \sin(314 \cdot t) \rightarrow \bar{V}_{an} = 300 \angle 0^\circ \text{ V (peak-value convention)}$

Similarly

$\bar{V}_{bn} = 250 \angle -90^\circ \text{ V}$

$\bar{V}_{cn} = 200 \angle 120^\circ \text{ V}$

$\bar{V}_{ca} = \bar{V}_{cn} - \bar{V}_{an} = 430 \angle 156^\circ \text{ V}$

$\bar{V}_{ab} = \bar{V}_{an} - \bar{V}_{bn}$
 $= 385 \angle 39.8^\circ \text{ V}$

$\bar{V}_{bc} = \bar{V}_{bn} - \bar{V}_{cn}$
 $= 420 \angle -77^\circ \text{ V}$

$v_{ab} = 385 \sin(314t + 39.8^\circ) \text{ V}$

$v_{bc} = 420 \sin(314t - 77^\circ) \text{ V}$

$v_{ca} = 430 \sin(314t + 156^\circ)$

2-marks for the correct approach if final answers are incorrect.

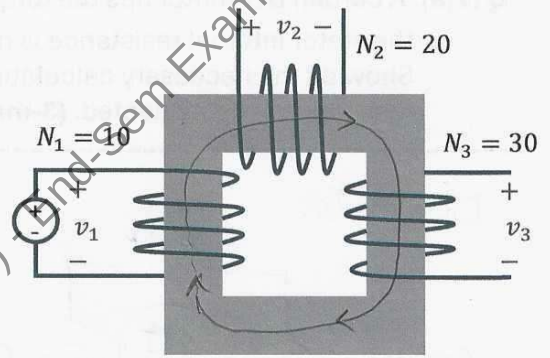
Analytical approach is also fine. Phasor diagram-based approach is only hinted, not mandatory.

2-marks each for correct final expression time-domain.

Allow variation in the answer up to 10%.

give 1-mark if answers are correct in phasor notation but not shown in time-domain expression

Q11(a): Following figure shows the schematic of a multi-winding transformer, where all the three windings are wound on a single core. The number of turns in a particular winding are shown next to that winding. If $v_1 = 2 \sin(314 \cdot t)$, sketch the waveform of v_1 , v_2 and v_3 in a single plot with common time-axis. Label the axes neatly denoting the key time-instances and minimum/maximum values. **(5-marks)**



$v_1 \rightarrow \Phi$ (flux in the core) $\rightarrow v_2$ & v_3 .

We know that $\frac{v_2}{v_1} = \frac{N_2}{N_1} = 2$ & $\frac{v_3}{v_1} = \frac{N_3}{N_1} = 3$,

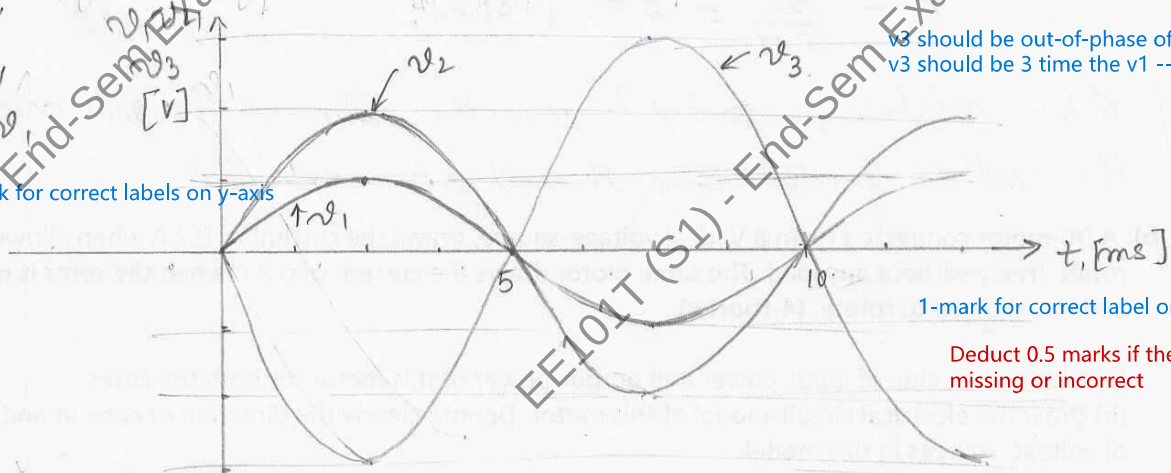
however, the polarity of the voltage is determined by the coil orientation on the core, determined by right-hand thumb-rule.

we get,

$$v_2 = 2v_1$$

$$\& v_3 = -3v_1$$

1-mark for correct labels on y-axis

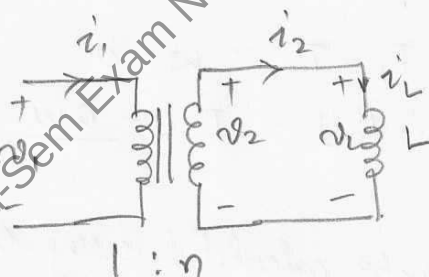


1-mark for correct label on time-axis

Deduct 0.5 marks if the unit of time is missing or incorrect

v_2 should be in phase with $v_1 \rightarrow$ 1-mark
 v_2 should be twice of $v_1 \rightarrow$ 0.5 marks
 v_3 should be out-of-phase of $v_1 \rightarrow$ 1-mark
 v_3 should be 3 times the $v_1 \rightarrow$ 0.5 marks

Q11(b): A transformer has turns-ratio of $1:n$ from primary to secondary terminals. Prove that an inductor of value L connected at the secondary terminals of this transformer appears like an inductor of value $\frac{L}{n^2}$ from the primary terminals. **(3-marks)**



1-mark for correct approach

$$v_L = L \frac{di_L}{dt}$$

we are interested in obtaining relation between v_1 & i_1

$$v_L = v_2, \quad i_L = i_2$$

$$v_2 = L \frac{di_2}{dt}$$

for this transformer,

$$v_2 = n v_1$$

$$\& i_2 = \frac{i_1}{n}$$

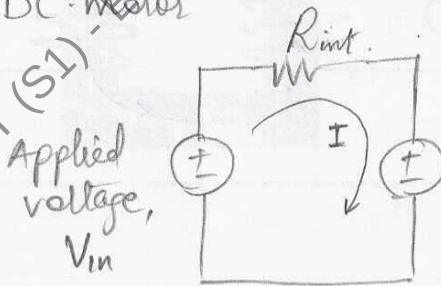
$$v_1 = \left(\frac{L}{n^2} \right) \frac{di_1}{dt} \Rightarrow v_1 = L' \frac{di_1}{dt} \text{ where } L' = \frac{L}{n^2}$$

1-mark for arriving at this equation

1-mark for explaining from primary terminal point of view.

Q12(a): A certain DC-motor has the torque-constant of $2 \frac{\text{N}\cdot\text{m}}{\text{A}}$, and the efficiency of 85%. Assuming that the stator internal resistance is negligible, obtain the value of back-emf constant of this motor. Show all the necessary calculation. Comment on the value of this constant if the stator internal resistance is no neglected. **(3-marks)**

DC-motor



if R_{int} is neglected, then

$$E_b = V_{in}$$

$$E_b : \text{back-emf} = k_b \cdot \omega$$

↪ back-emf constant.

$$\therefore \text{Input power } P_{in} = V_{in} \cdot I = E_b \cdot I$$

$$\text{Now, } P_{out} = T \cdot \omega = (k_t \cdot I) \omega, \text{ But } P_{out} = \eta \cdot P_{in}$$

1-mark for correct approach

$$(k_t \cdot I) \omega = \eta \cdot (E_b \cdot I) = \eta (k_b \cdot \omega) \cdot I \Rightarrow k_b = \frac{k_t}{\eta}$$

$$\therefore k_b = \frac{2}{0.85} = 2.35 \text{ [V}\cdot\text{s/rad]}$$

1-mark for correct value of k_b

Deduct 0.5 marks if the unit of back-emf constant is missing or incorrect

Even if stator resistance is included, then the value of k_b will not change, only the method for obtaining it will get complicated. 1-mark for correct response.

Q12(b): A DC-motor connected to an 8 V ideal voltage-source, draws the current of 0.2 A when allowed to rotate freely without any load. The same motor draws the current of 0.8 A when the rotor is blocked and not allowed to rotate. **(4-marks)**

(a) Obtain the value of input-power and output-power of this motor for both the cases.

(b) Draw the electrical circuit-model of this motor. Denote clearly the direction of current and polarities of voltage-sources in this model.

(a) Case 1: 8V & 0.2A → No-load, & rotating

$$P_{in} = 8 \times 0.2 = 1.6 \text{ W}$$

0.5 marks for correct input value

$$T_L = 0$$

$$\therefore P_{out} = 0 \cdot \omega = 0 \text{ W}$$

0.5 marks for correct output value

Case 2: 8V & 0.8A →

$$P_{in} = 8 \times 0.8 = 6.4 \text{ W}$$

0.5 marks for correct input value

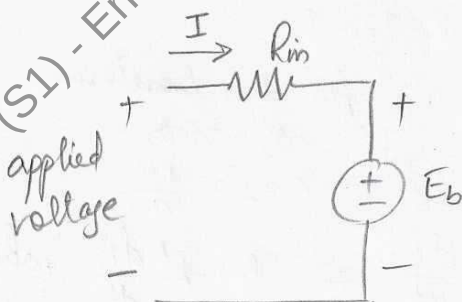
$$T_L = T \text{ & } \omega = 0$$

$$\therefore P_{out} = T \cdot \omega = 0 \text{ W}$$

0.5 marks for correct output value

Deduct 0.5 marks if the unit of power is missing or incorrect

(b)



1-mark for correct model

R_{int} can be calculated using case 2. as follows $R_{int} = 8/0.8 = 10 \Omega$

& E_b using case 1:

$$E_b = 8 \text{ V} - 0.2 \times 10 = 6 \text{ V}$$

0.5 marks correct R_{int} and 0.5 marks for correct E_b